GYMFLOW GYM MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

Rohith D

(2116240701435)

in partial fulfilment of the End Semester Examination for the course

CB23F35

LOW CODE APPLICATION DEVELOPMENT
USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu - 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "GYMFLOW GYM MANAGEMENT SYSTEM" is the bonafide work of Rohith D (2116240701435), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 - Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, Rohith D (2116240701435), declare that the project report titled "GYMFLOW GYM MANAGEMENT SYSTEM", submitted for the End Semester Examination of CB23F35 - Low-Code Application Development using Zoho Creator, describes my individual course project. The application combines linked forms, reports, pages, Deluge workflows and an owner email notification in Zoho Creator.

Platform documentation and AI assistance used during development are acknowledged. Zia assisted with design suggestions and generation of the email function; an AI coding assistant supported implementation and report preparation. Generated logic was reviewed against the application schema and tested. The report distinguishes completed functionality from work that remains to be verified.

Place: Chennai
Date: _____

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ACKNOWLEDGEMENT

I thank the management and Principal of Rajalakshmi Engineering College for providing the facilities and learning environment for this project. I express my gratitude to Dr. MANORANJINI J, Head of the Department of Computer Science and Engineering, for departmental support.

I sincerely thank my course faculty, Dr.Jinu Sophia, for guidance during the course. I acknowledge Zoho Creator and its documentation for the application platform, Deluge reference and development tools. I also thank my family and friends for their support.

Rohith D

ABSTRACT

GymFlow is a gym management application developed on Zoho Creator to connect membership administration, payment records, attendance, class scheduling and operating expenses. A gym that uses separate registers or spreadsheets must repeatedly reconcile member information, subscription periods and receipts before making operational decisions. GymFlow addresses this problem through a shared data model and validation at the point of entry. The application contains nine related forms: Plans, Trainers, Members, Memberships, Payments, Attendance, Classes, Bookings and Expenses. Eight lookup relationships connect these records, while nine list reports support review and editing. Deluge workflows calculate membership expiry and discounted fees, prevent overlapping memberships, reject overpayments and duplicate open visits, and check booking eligibility and class capacity. The main Dashboard presents six colored indicators and four charts calculated from saved records. A second page, Member Home, contains email-based member filtering, although the portal permission boundary and external member sign-in still require verification. An owner-only function sends a current dashboard summary through the Creator sendmail task; delivery to the college inbox was confirmed. Zia assisted with design review and generated the initial email function, which was corrected and tested before use. Verification used 45 demonstration records, with five records in each form. The observed November totals were INR 8,600 in collections, INR 6,000 in expenses, INR 2,600 in net cash and INR 3,650 in unpaid membership balances. The work demonstrates how visual components and focused code can produce a connected operational application. It also identifies the remaining portal, concurrency and deployment checks required before use with real members.

Keywords: Gym management; Zoho Creator; Low-code development; Deluge; Dashboard; Email notification.

PROJECT AT A GLANCE

Table P1 Project profile

Attribute	GymFlow implementation
Project title	GymFlow Gym Management System
Student	Rohith D | 2116240701435
Department	Computer Science and Engineering
Course faculty	Dr.Jinu Sophia
Course	CB23F35 - Low-Code Application Development using Zoho Creator
Platform and account	Zoho Creator | rohith.d.2024.cse@rajalakshmi.edu.in
Forms and data fields	9 forms; 48 user data fields; 8 lookup fields; 0 subforms
Reports and pages	9 list reports; Dashboard and Member Home; 4 custom charts
Workflows and functions	Form validations across the 9 forms; membership defaults; 1 standalone Deluge function
Automation not used	No scheduled, approval, Blueprint or batch workflow
Integration	Native Creator sendmail; owner summary delivered and confirmed
Demonstration dataset	45 records; 5 records per form
Access status	Owner/staff demonstration verified; member portal isolation and external sign-in pending
Environment	Single existing Creator application; no staged production promotion demonstrated
Verification date	9 November 2026

Live application

https://creatorapp.zoho.in/rohith.d.2024.cse_rajalak93/gymflow/#Page:Dashboard

Demonstration access uses the existing owner account. Passwords and verification codes are not included in this report.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create Read Update Delete
CSS	Cascading Style Sheets
ER	Entity Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
INR	Indian Rupee
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
UPI	Unified Payments Interface
ZML	Zoho Markup Language

Deluge is the scripting language used for application rules and the email function. Zia is the AI development assistant used in this project.

CHAPTER 1

INTRODUCTION

1.1 Background

A gym manages people, subscription periods, money, visits and scheduled training sessions. These activities are connected: a class booking depends on membership validity, and an outstanding balance depends on the membership fee and all related payments. Keeping each activity in a separate file makes these dependencies difficult to maintain.

GymFlow is an academic demonstration application for this operational setting. It uses Zoho Creator as the managed application and data platform. The implementation is based on the project requirements and demonstration records; it does not claim a field study of a named commercial gym.

1.2 Problem Statement

The problem is to maintain a consistent view of gym operations while preventing invalid or duplicate transactions. Staff need to find a member, issue a valid membership, record partial payments, track visits and manage class capacity without repeatedly copying the same details. Management needs current collections, expenses and balances from those same records.

• Separate records can repeat member details and disagree about subscription dates.

• Manual fee and balance calculations can admit overpayments or incorrect discounts.

• Uncontrolled visits and bookings can create duplicate open attendance or duplicate reservations.

• A manually compiled summary can become outdated as soon as records change.

1.3 Objectives

• Connect nine gym activities through eight lookup relationships and maintain five demonstration records per form.

• Calculate membership expiry and total fees, and validate payment, attendance and booking rules.

• Provide a dashboard with six indicators and four charts based on saved records.

• Send an owner summary email and verify receipt in the college inbox.

• Document testing evidence, AI assistance and the remaining access-control limitations.

1.4 Scope of the Project

The implemented scope covers plans, trainers, members, memberships, payments, attendance, classes, bookings, expenses, staff reports and owner notifications. Money fields record amounts in INR. Recording a payment does not initiate a bank transfer or card charge. Equipment inventory, workout prescriptions, biometric attendance and payment-gateway processing are outside this version.

1.5 Justification for a Low Code Approach

A low-code platform supplies reusable data and interface components while allowing business-specific rules to be expressed in code [1]. In GymFlow, the visual form and report builders reduce the need to implement routine database screens. Deluge provides the checks that visual field configuration alone cannot express. This combination suits a compact operational application while keeping the schema and rules visible to its maintainer.

1.6 Organisation of the Report

Chapter 2 reviews the system context and related sources. Chapters 3 and 4 describe requirements and design. Chapter 5 presents implementation evidence. Chapters 6 and 7 cover testing, deployment and security. Chapters 8 and 9 discuss results, limitations and future work. The appendices contain feature coverage, course mapping, the data dictionary, source extracts, user instructions and examination records.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The comparison baseline is a conventional manual process. A receptionist records member details, chooses a plan, writes subscription dates and records receipts separately. Visits are entered in a register and classes are maintained in a schedule. Expenses are collected in another sheet. A summary therefore requires matching names and dates across several records.

2.2 Limitations of the Existing System

This baseline offers limited automatic validation and relies on the operator to detect duplication. Renewals can overwrite previous subscription information, multiple receipts can be missed when calculating a balance, and capacity checks can be overlooked. These are design risks of the baseline, rather than measured incidents at a particular gym.

2.3 Low Code versus Traditional Development

Table 2.1 Development approach comparison

Aspect	Traditional custom application	GymFlow on Zoho Creator
Data entry	Build interface and persistence code	Configure forms and list reports
Rules	Implement server handlers and validation	Use Deluge form workflows
Hosting	Configure and maintain application services	Use the Creator hosted application
Customization	Full control over stack and UI	Custom pages within Creator limits
Maintenance	Maintain framework and application code	Maintain schema, rules and plan settings
Evidence of effort	No comparison build was timed	No numerical time-saving claim is made

2.4 Literature Survey

Table 2.2 Sources and relevance

Source	Relevant finding	Application to GymFlow
[1] Bock and Frank, 2021	Low-code platforms combine high-level models, UI tools and programmable behavior.	Separates platform capabilities from application-specific rules.

Source	Relevant finding	Application to GymFlow
[2] Zoho form builder documentation	Fields, form layout, defaults and validation are configurable.	Guides the nine forms and the treatment of required fields.
[3] Zoho gym application overview	Memberships, scheduling and financial administration are common gym functions.	Provides a domain comparison; the supplied vendor template was not deployed.
[4] Deluge sendmail documentation	A script can send a message to configured recipients.	Supports the owner summary function.
[5] Build Agent documentation	AI proposes application changes for review and application.	Explains the supervised use of Zia.
[6] Portal user documentation	External portal identities have a domain restriction.	Explains the separate member verification test.

2.5 Emerging Trends in Low Code Development

AI assistance can turn a natural-language requirement into a draft component, but the generated field names and conditions still need review. GymFlow illustrates this requirement: Zia initially used an Active membership-status check, whereas the application stores Valid and Cancelled. The condition was corrected and date coverage was added before the email function was tested.

2.6 Proposed System

GymFlow replaces disconnected records with linked forms and shared calculations. Member details are stored once, memberships preserve their purchase period and fee, and payments refer to the appropriate membership. Attendance and bookings use membership eligibility. Dashboard queries summarize the resulting records, and the owner email uses the same date boundaries and totals.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1 Stakeholders and access status

Stakeholder	Purpose	Status in this project
Owner or administrator	Configure application, manage gym records, view totals and send summary	Owner account demonstrated
Front desk staff	Register members, record payments and visits, manage bookings	Operational workflows demonstrated through owner session; separate staff identity not tested
Trainer	Use class and member assignment information	Trainer records exist; separate trainer login not implemented
Member	View own subscription and related information	Member Home developed; portal restrictions and sign-in not yet verified

3.2 Functional Requirements

Table 3.2 Functional requirements

ID	Requirement	Priority
FR01	Maintain plans, trainers and member profiles with required fields.	High
FR02	Link members and plans to memberships and calculate inclusive expiry and discounted total.	High
FR03	Record partial or full payments without exceeding the membership balance.	High
FR04	Record valid check-in and check-out times and reject duplicate open visits.	High
FR05	Schedule classes and validate trainers, times and capacity.	High
FR06	Create eligible, nonduplicate class bookings.	High
FR07	Record expenses and produce six indicators and four charts.	High
FR08	Send a current summary to the owner email address.	Medium
FR09	Restrict external members to their own records.	High; pending verification

3.3 Non Functional Requirements

Table 3.3 Non functional requirements

Category	Requirement and evidence
Usability	Dashboard opens first; navigation and quick actions lead to existing forms and reports.
Integrity	Validation runs before saving relevant transactions; lookup fields link existing records.
Security	Owner email function has a login guard. Portal isolation must be completed before real-member rollout.
Maintainability	Separate forms, named workflows and saved scripts make individual rules inspectable.
Performance	Demonstration uses 45 records. No load benchmark or response-time guarantee is claimed.
Availability	Requires working internet, a valid Zoho session and an active Creator plan.

3.4 Hardware and Software Requirements

Table 3.4 Execution environment

Item	Requirement
Application platform	Zoho Creator web application in the India data center
Programming	Deluge; HTML and CSS page snippet; ZML button
Client	A computer with a supported modern web browser and internet connection
Email	Owner college inbox for summary delivery; external-domain inbox for a future portal test
Test dataset	Five related demonstration records in each of nine forms
Optional tools not used	No separate backend server, external database, payment API or API test service is required for this version

3.5 Use Case Model

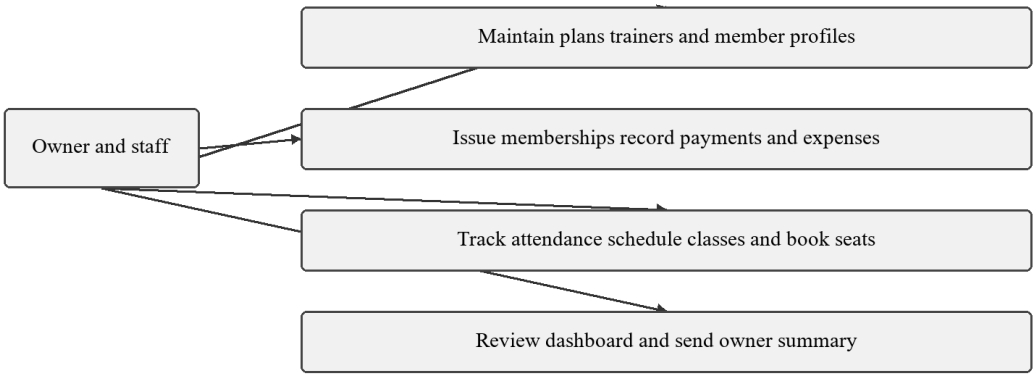


Figure 3.1 Implemented owner and staff use cases

Table 3.5 Key use cases

Use case	Precondition and flow	Result
UC01 Issue membership	An active member and plan exist. Select both, enter start date and discount, validate eligibility and overlap, then save.	Expiry, fee snapshot, discounted total and readable reference are stored.
UC02 Record payment	A Valid membership exists. Select its reference and enter date, amount and method. Compare amount with remaining balance.	Valid payment saved; invalid amount rejected.
UC03 Send owner summary	Owner is signed in. Click the Dashboard email button once. Fetch current totals and send to owner.	Submission log displayed; college inbox receipt confirmed.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

GymFlow runs within Zoho Creator. The presentation layer contains native forms and reports plus two custom pages. Deluge workflows enforce rules on submitted records. The data layer consists of nine forms connected by lookup fields. The integration layer is the native sendmail task, which delivers the owner summary. No separate application server or external REST endpoint is part of this implementation.

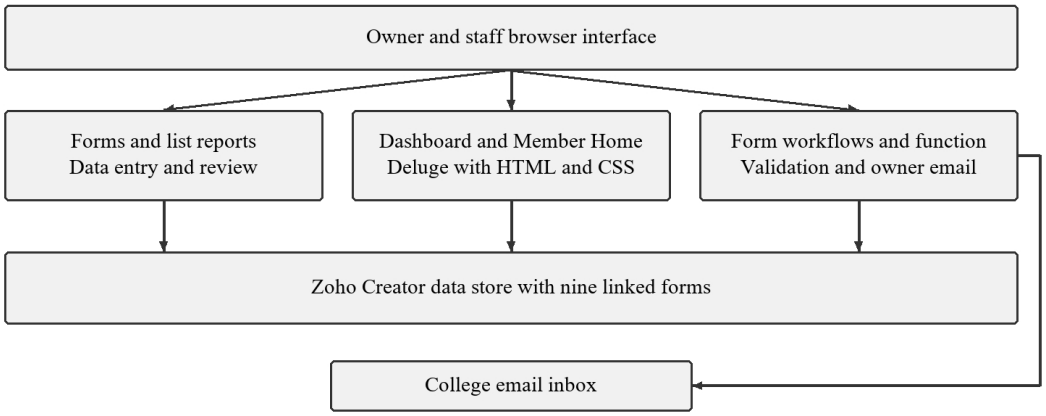


Figure 4.1 GymFlow application architecture

4.2 Data Model

A member profile is distinct from a membership purchase. This allows renewals without replacing the person record. Payments belong to a specific membership, while attendance belongs to a member. Bookings form the association between members and classes. Trainers can be referenced by member assignments and class schedules. Expenses are independent operating records.

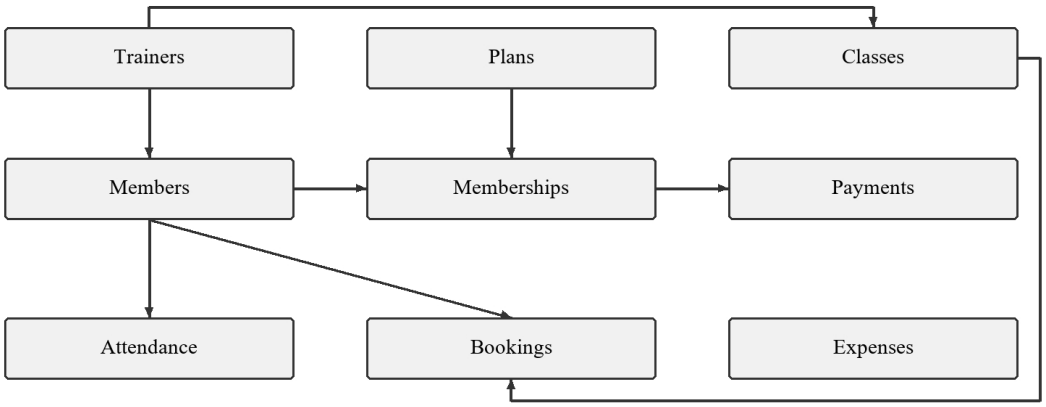


Figure 4.2 Relationships between the nine GymFlow forms

Table 4.1 Lookup relationships

Source field	Target	Relationship
Members.Assigned_Trainer	Trainers	Many members to one trainer
Memberships.Member	Members	Many membership periods to one member
Memberships.Plan	Plans	Many memberships to one plan
Payments.Membership	Memberships	Many payments to one membership
Attendance.Member	Members	Many visits to one member
Classes.Trainer	Trainers	Many classes to one trainer
Bookings.Member	Members	Many bookings to one member
Bookings.Gym_Class	Classes	Many bookings to one class

4.3 Form Design

The application has 48 user data fields across nine forms. It uses Single Line, Multi Line, Number, Decimal, Email, Date, Date-Time, Drop Down and Lookup field categories. The phone values are stored as text, and monetary amounts use Decimal fields labelled INR. They are not native Phone or Currency fields. No subform is used because visits, payments and bookings need independent transaction records. Appendix C lists the actual field names, types and mandatory status.

Table 4.2 Form inventory

Form	Fields	Main purpose
Plans	5	Reusable package duration and fee
Trainers	5	Trainer contact details and active status
Members	6	Member profile and trainer assignment
Memberships	9	Subscription period, price and discount
Payments	6	Membership receipts and payment methods
Attendance	3	Member visit timestamps
Classes	6	Trainer sessions and capacity
Bookings	3	Member reservations for classes
Expenses	5	Operating cost records

4.4 Process and Workflow Design

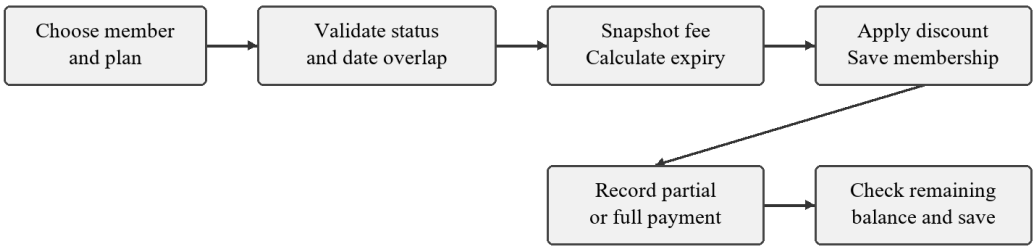


Figure 4.3 Membership and payment validation flow

Membership creation snapshots the plan price and calculates End Date as Start Date plus Duration Days minus one. Discount is checked against the fee, and Total is Fee minus Discount. The workflow rejects overlapping Valid memberships. Existing member, plan and start date cannot be changed during an edit; a renewal requires a new membership. Payment checks exclude the payment currently being edited when summing earlier receipts.

4.5 User Interface and Navigation Design

Dashboard is the first section and landing page. Six cards distinguish active memberships, open visits, collections, expenses, net cash and unpaid balances. Four charts follow the cards. Front-desk quick actions link to membership, payment, check-in and booking forms and their reports. The nine forms remain separate because they represent different data entities and transactions.

4.6 Access Control Design

Table 4.3 Intended access and verification

Component	Owner	External member target	Verified status
Gym forms and staff reports	Manage	No access	Owner access demonstrated
Dashboard	View and email	No access	Owner dashboard demonstrated
Member Home	Inspect page	View own records	Page filtering only; permission changes did not persist; two-identity tests pending
Send_Demo_Summary	Execute	Denied by function guard	Owner execution verified; separate negative identity test pending

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The GymFlow application is hosted in Rohith D's Zoho Creator account. Its navigation contains only gym-related modules, with Dashboard first. Figure 5.1 shows the live application with the account profile open. Development uses native Creator components and custom Deluge rules rather than the sample farm application or vendor gym template.

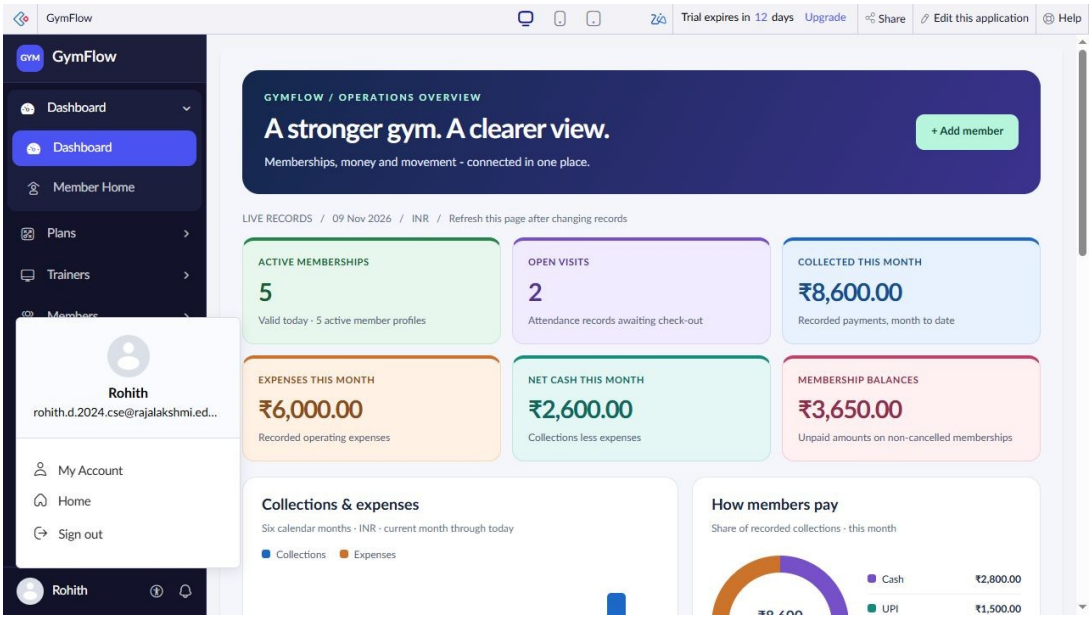


Figure 5.1 GymFlow dashboard with the Zoho account profile

Owner account: rohith.d.2024.cse@rajalakshmi.edu.in. The account profile and application name establish the source of the screenshots. The dataset is demonstration data retained for the project presentation.

5.2 Forms and Data Capture

Each form stores a distinct entity or transaction. Required fields are indicated in the live interface. Defaults and workflow checks are applied where needed; the following figures show the actual entry screens. Separate list reports expose the saved records.

5.2.1 Plans

The Plans form stores the package name, duration, fee, status and benefits. Positive duration, nonnegative fees and unique plan names support reusable subscription packages.

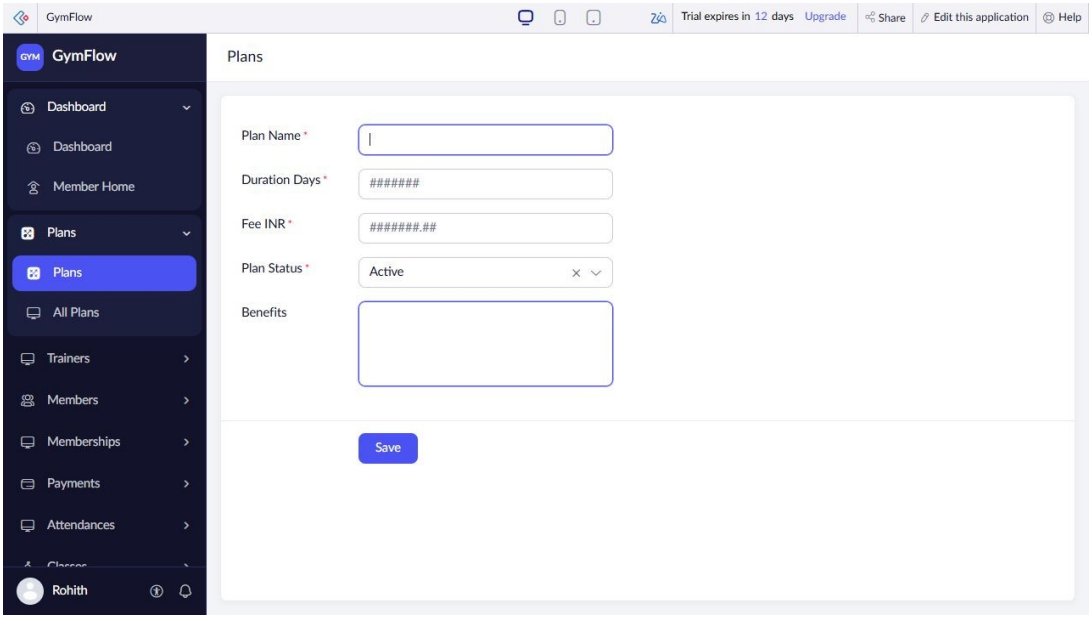


Figure 5.2 Plans form in live mode

5.2.2 Trainers

The Trainers form stores name, email, optional phone and specialization, and active status. Email is normalized and checked for duplication before saving.

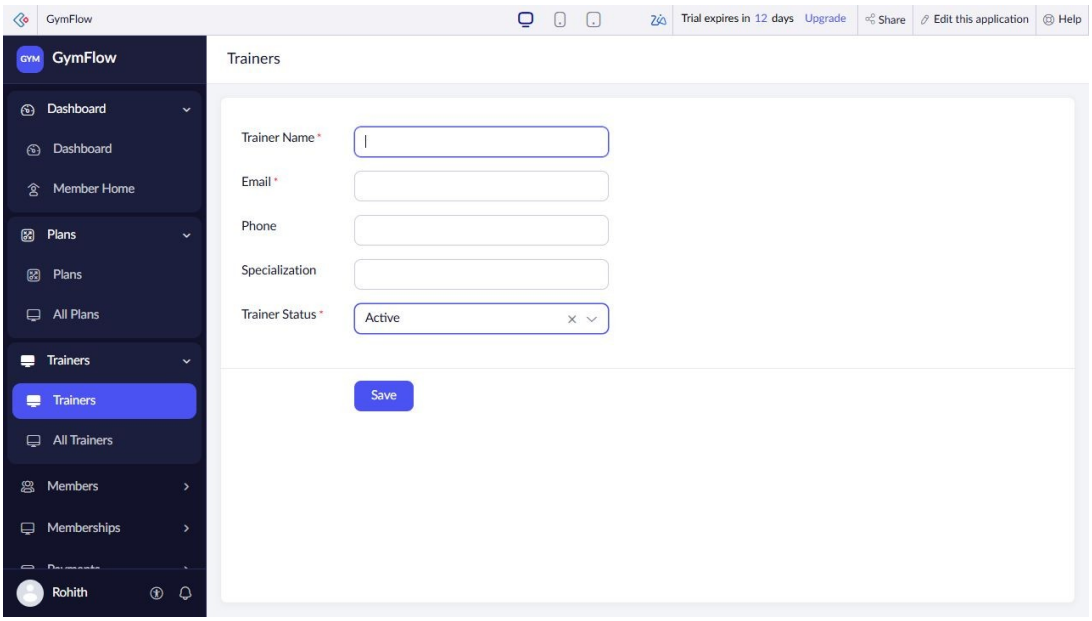


Figure 5.3 Trainers form in live mode

5.2.3 Members

The Members form captures identity, email, joining date, optional phone and trainer assignment. Keeping the profile separate avoids repeating contact data on every transaction.

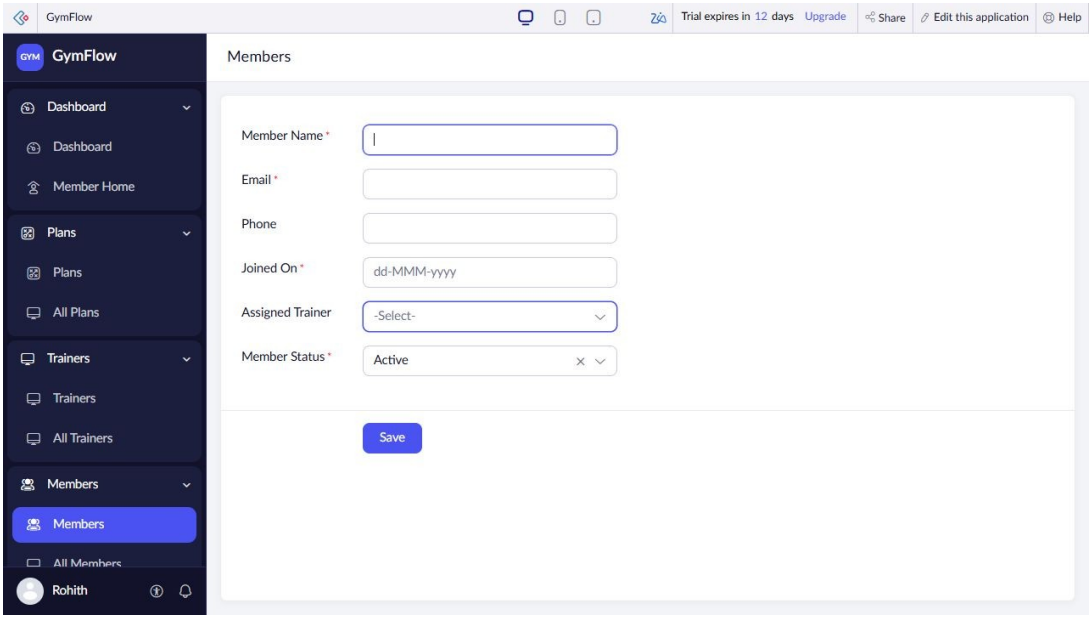


Figure 5.4 Members form in live mode

5.2.4 Memberships

The Memberships form links a member to a plan. End Date, Fee and Total are calculated and disabled for manual entry; Discount starts at zero. The readable membership reference is generated automatically.

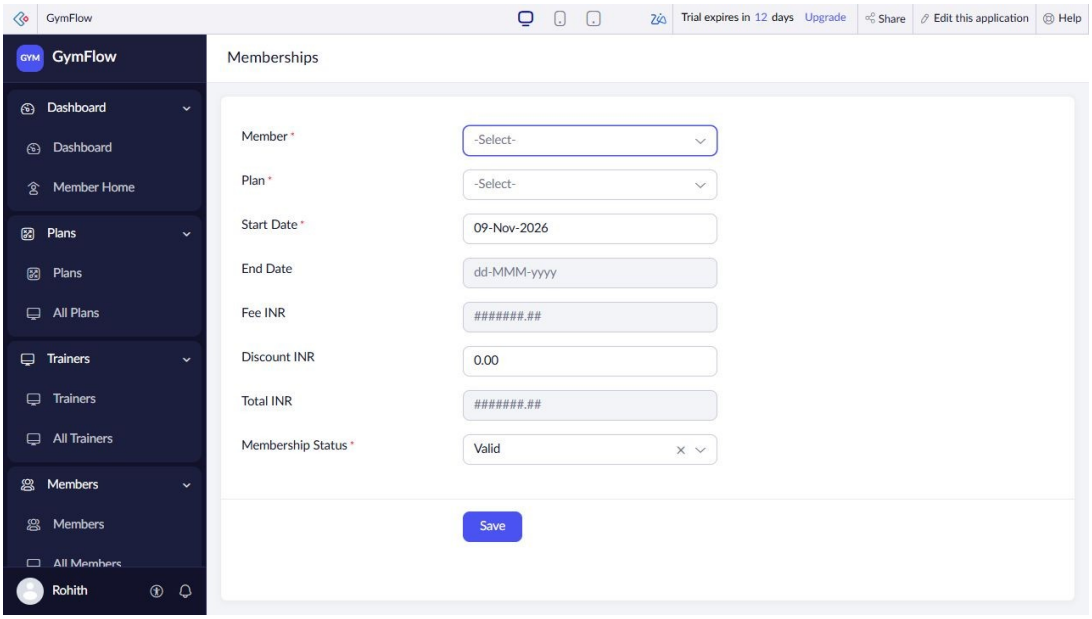


Figure 5.5 Memberships form in live mode

5.2.5 Payments

The Payments form selects the readable membership reference and stores date, amount, method, reference and notes. Its workflow prevents cancelled-membership payments, overpayment and future payment dates.

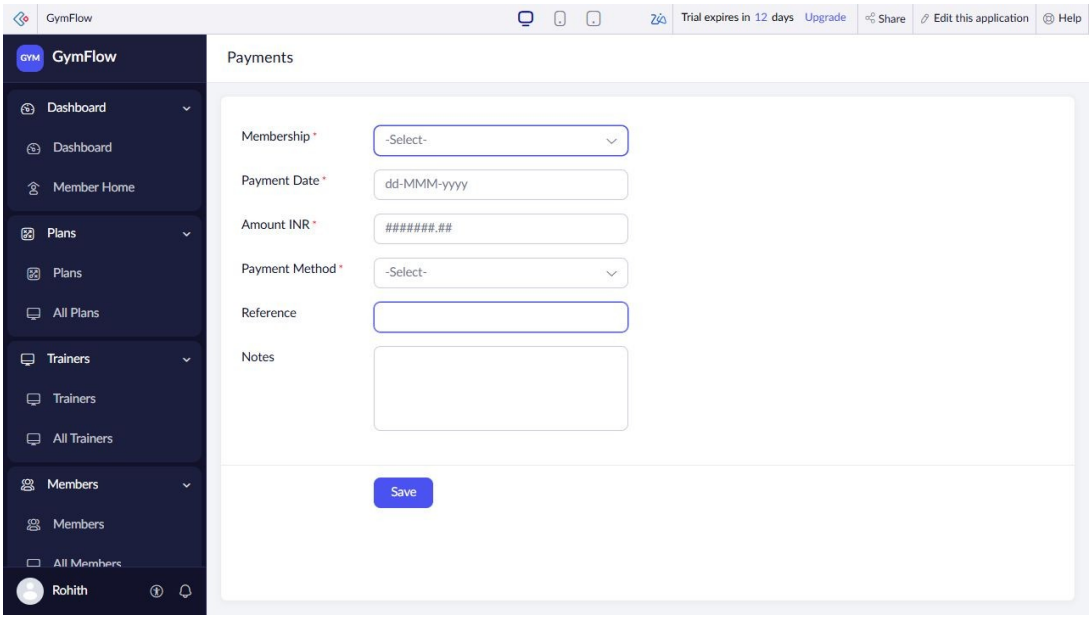


Figure 5.6 Payments form in live mode

5.2.6 Attendance

The Attendance form stores a member and check-in/check-out timestamps. A valid membership must cover the visit date. A second open visit and inconsistent timestamps are rejected.

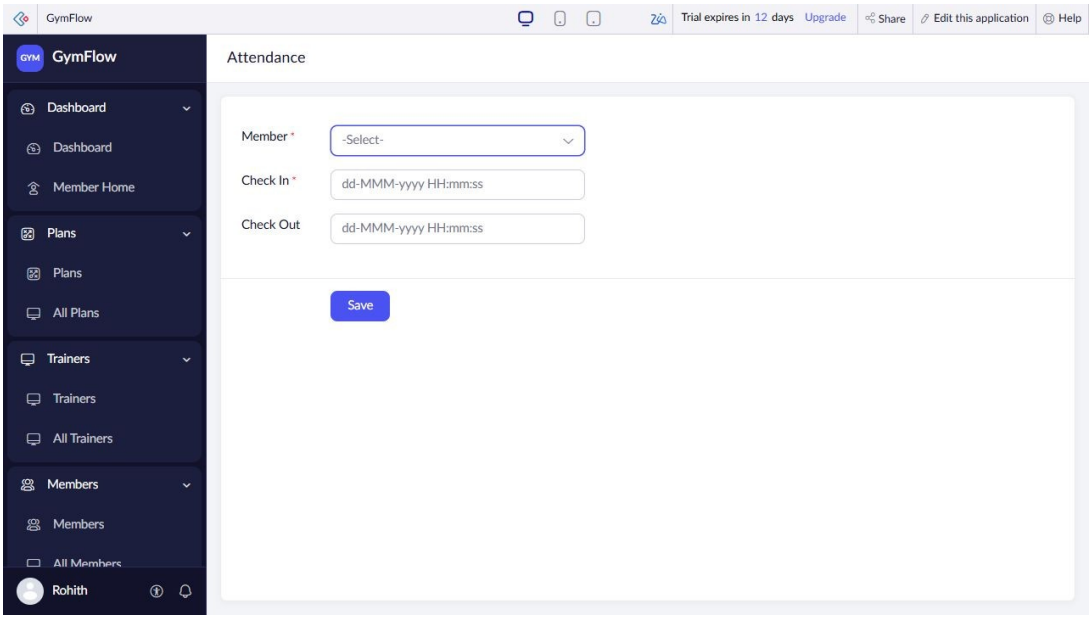


Figure 5.7 Attendance form in live mode

5.2.7 Classes

The Classes form records a title, trainer, start and end, capacity and status. The workflow checks an active trainer, a valid interval, schedule overlap and capacity changes.

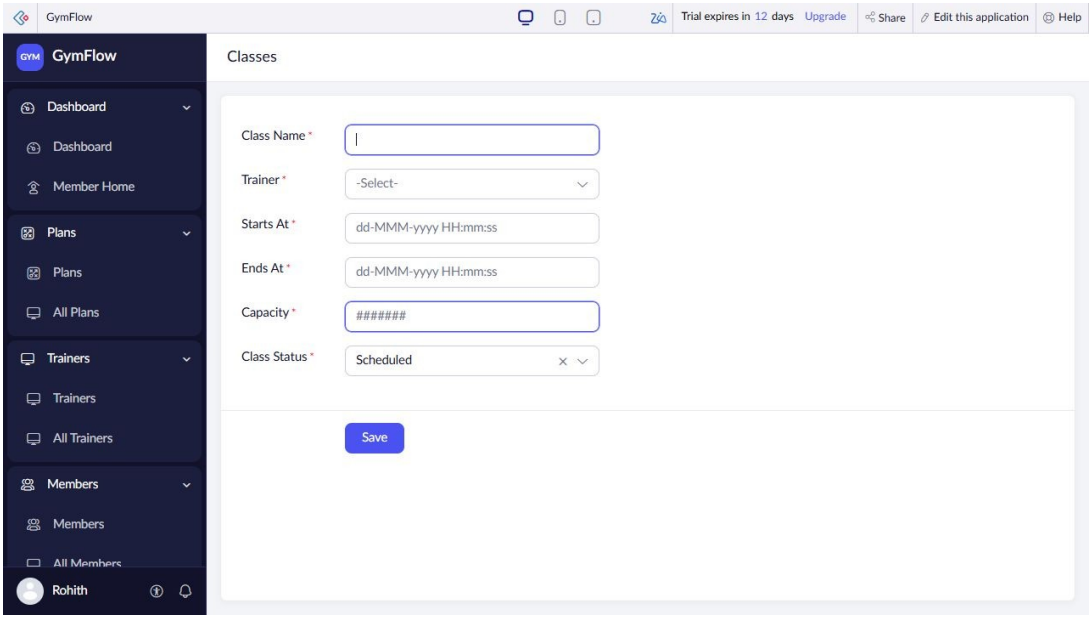


Figure 5.8 Classes form in live mode

5.2.8 Bookings

The Bookings form links a member to a class with a booking status. Eligibility, duplicate reservations, class availability and remaining capacity are checked before saving.

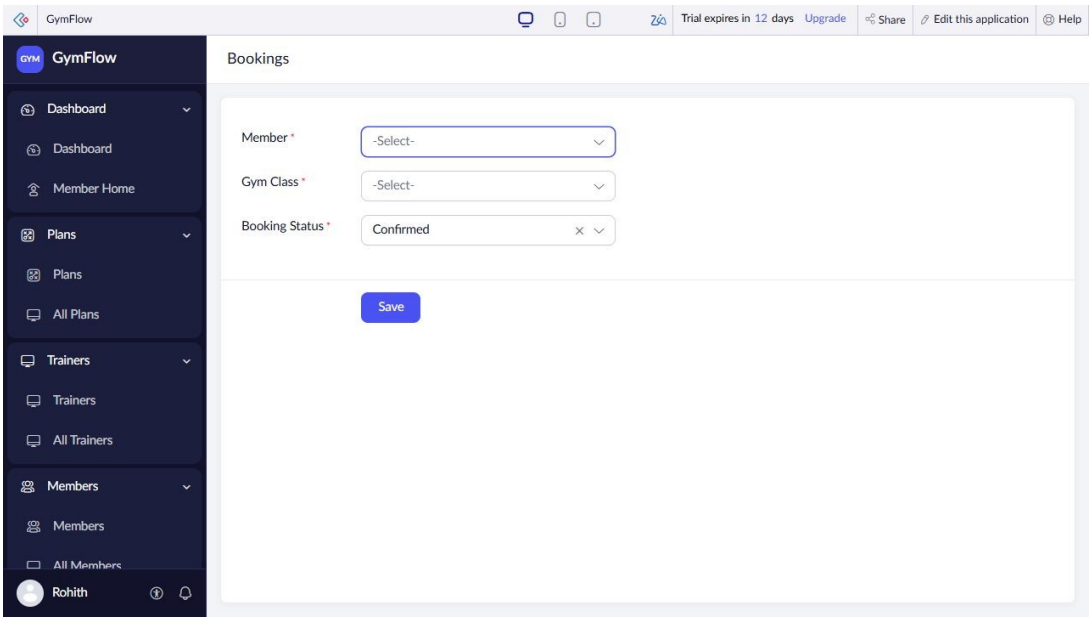


Figure 5.9 Bookings form in live mode

5.2.9 Expenses

The Expenses form records date, category, amount, description and reference. Positive expense amounts feed the monthly spending chart and net-cash calculation.

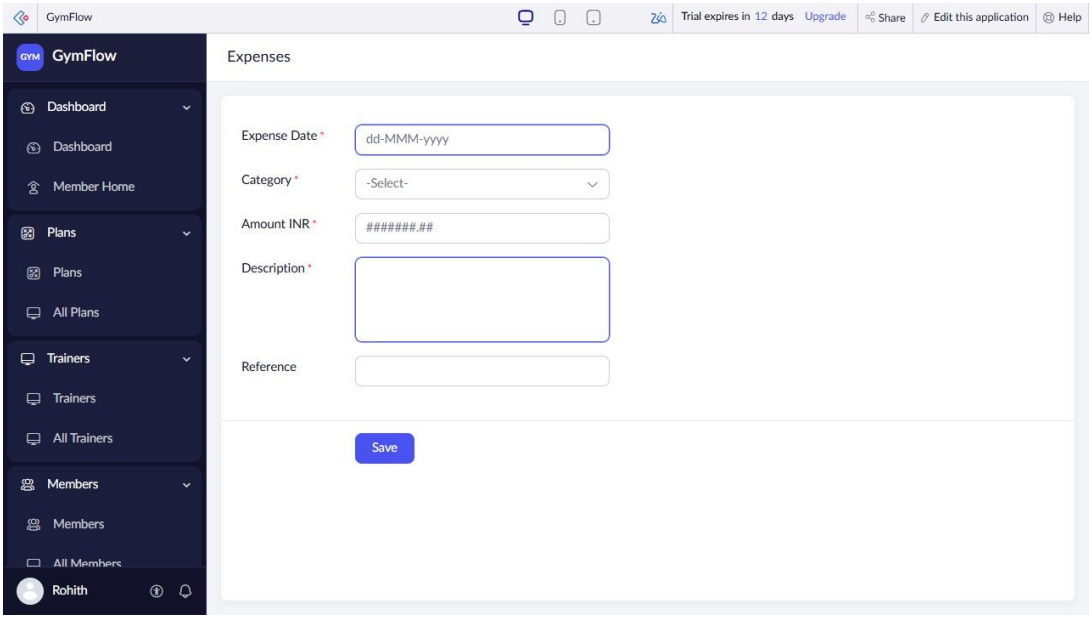


Figure 5.10 Expenses form in live mode

5.3 Reports

Nine native list reports provide the saved records. These reports are not Kanban, calendar or native pivot reports. The dashboard charts are separately rendered by the page snippet.

Table 5.1 Report inventory

Report	Source form	Operational use
All Plans	Plans	Review packages
All Trainers	Trainers	Review trainer profiles
All Members	Members	Review member profiles
All Memberships	Memberships	Review periods, totals and status
All Payments	Payments	Review receipts and methods
All Attendances	Attendance	Review visits and check-out
Classes Report	Classes	Review session schedule
All Bookings	Bookings	Review reservations
All Expenses	Expenses	Review operating costs

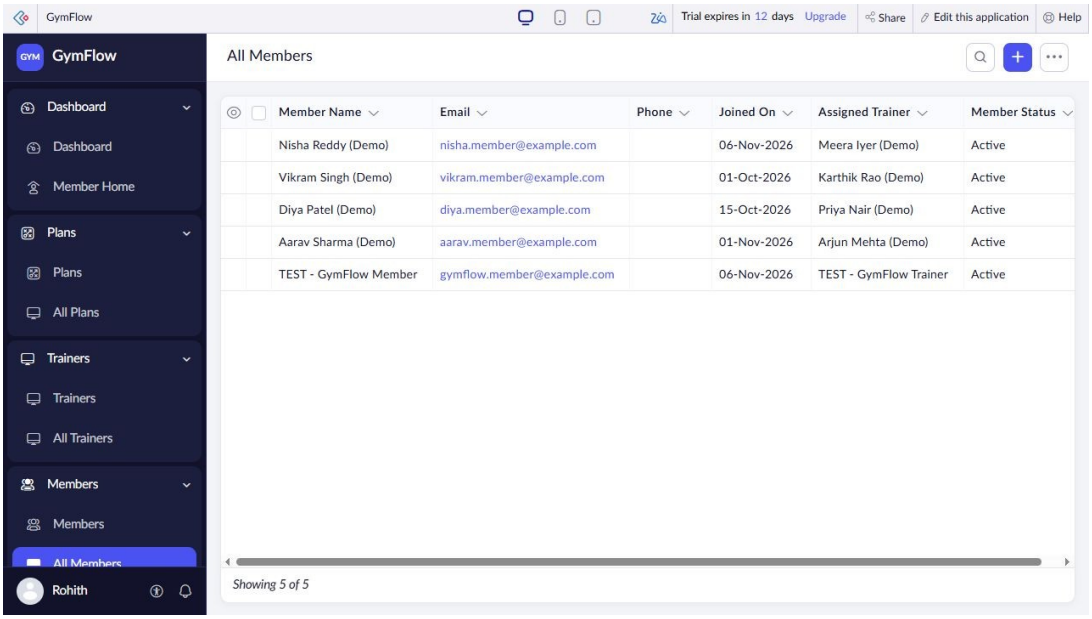


Figure 5.11 Member directory with five demonstration records

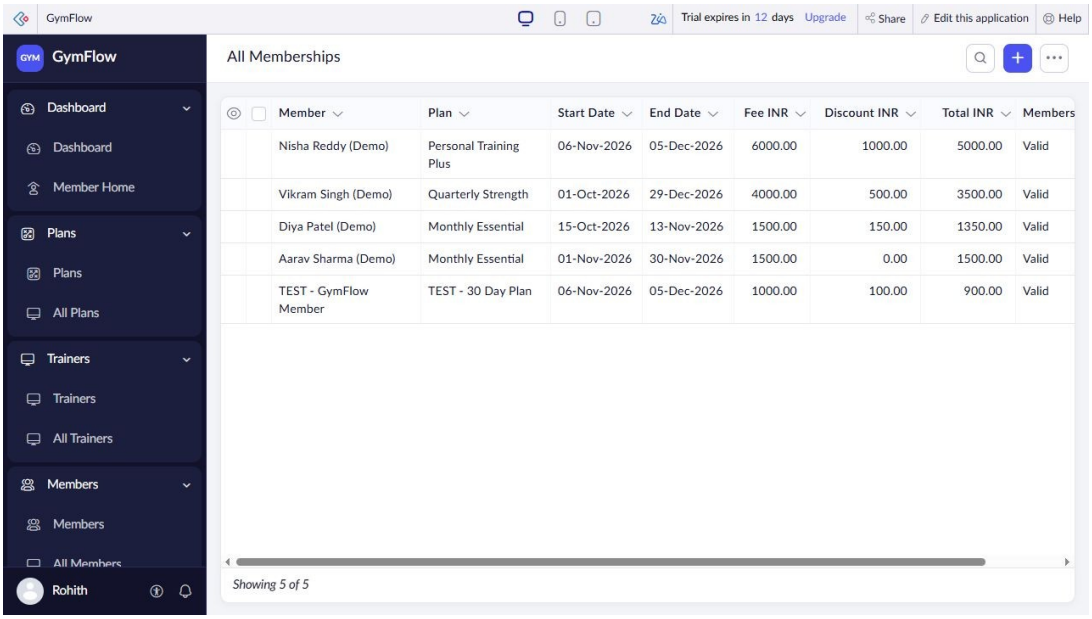


Figure 5.12 Membership periods and calculated amounts

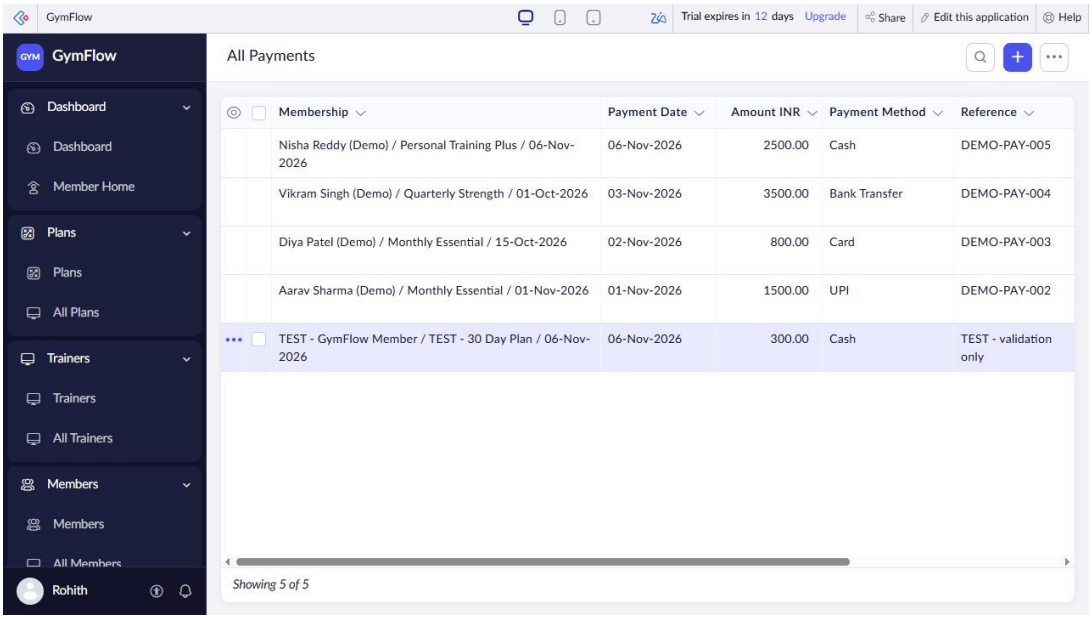


Figure 5.13 Payment history with five saved receipts

5.4 Pages and Dashboards

Dashboard uses Deluge queries in an HTML snippet to compute indicators and chart values. HTML and CSS render the cards, grouped columns, donut and horizontal bars. Figures recalculate on page load, so the page must be refreshed after a record change. The graphs are custom charts, not native Creator pivot charts.

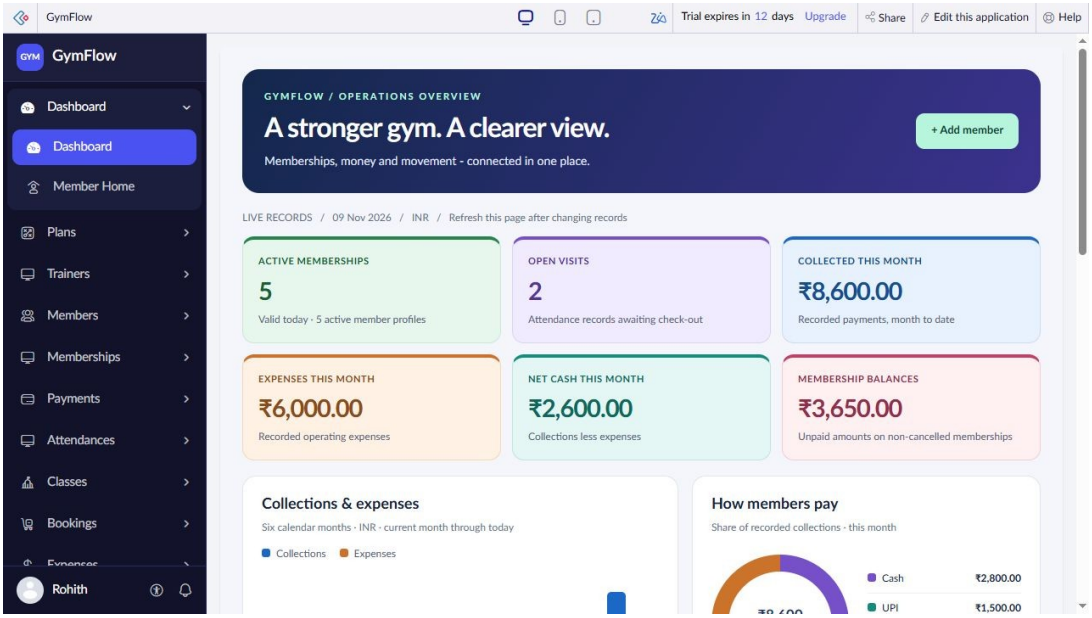


Figure 5.14 Main dashboard with six colored indicators

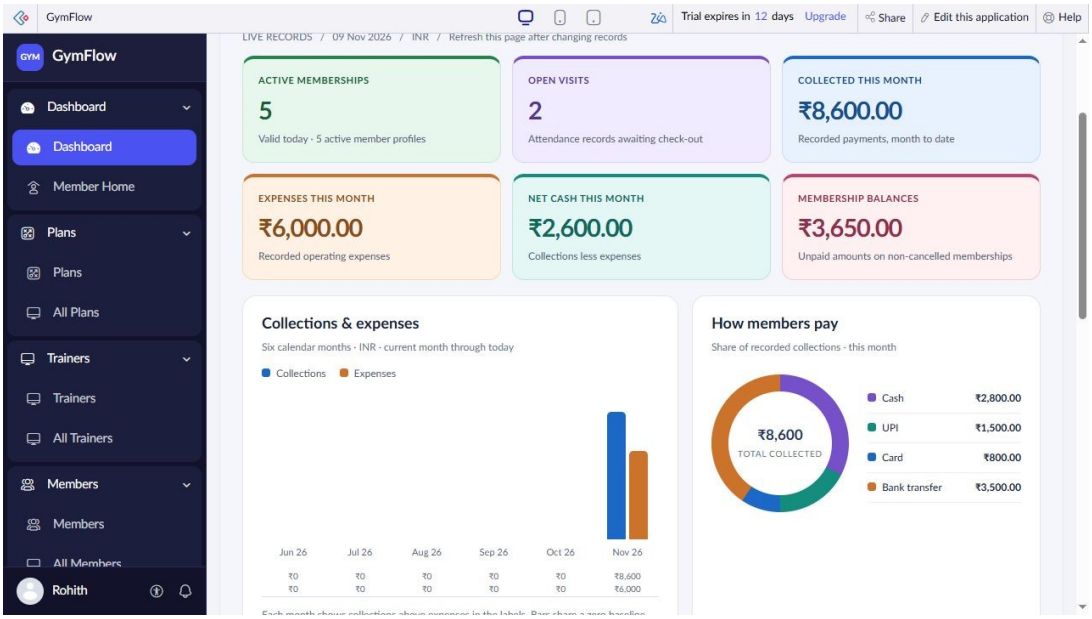


Figure 5.15 Collections expenses and payment method charts

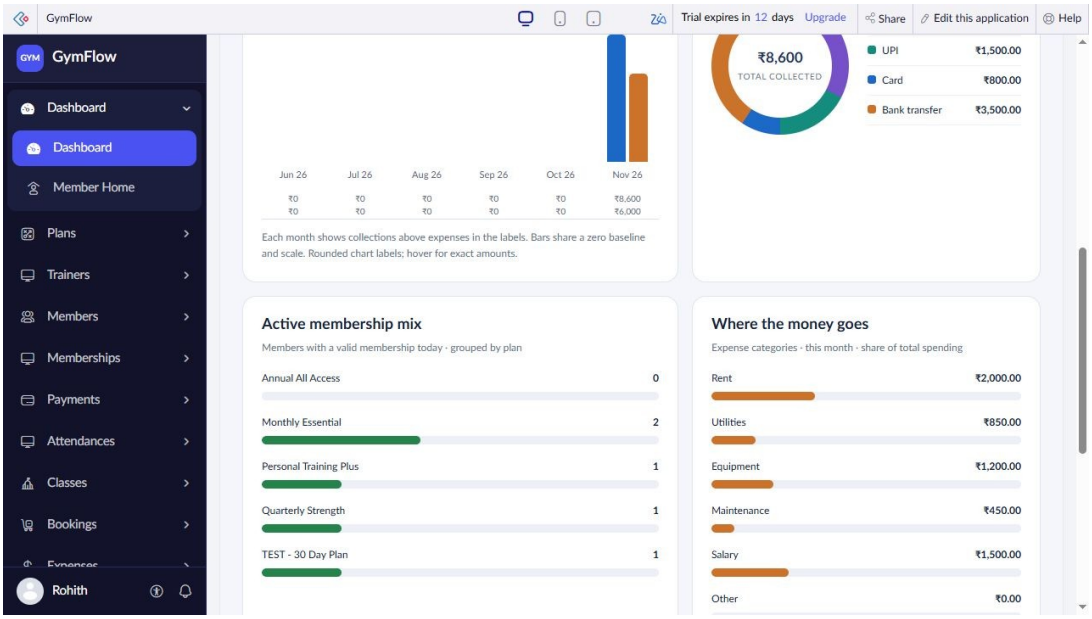


Figure 5.16 Active plan mix and expense category charts

Table 5.2 Dashboard calculations

Indicator or chart	Definition
Active memberships	Membership_Status is Valid and Start_Date <= today <= End_Date
Open visits	Attendance records without Check_Out; not necessarily visits today
Month collections	Payment amounts from first day of current month through today
Month expenses	Expense amounts for the same month-to-date interval
Net cash	Month collections minus month expenses
Membership balances	Unpaid amount on non-cancelled memberships
Six month comparison	Monthly payment and expense sums, retaining zero months

Indicator or chart	Definition
Payment mix	This-month collections grouped by payment method
Active plan mix	Valid memberships covering today grouped by plan
Expense categories	This-month spending grouped by category

5.5 Deluge Scripting

The scripts combine record fetches, aggregate sums and counts, conditional checks, date calculations, alerts and cancellation of invalid submissions. A null-safe sum treats no existing payments as zero. The email function reuses the current date and current month boundaries to match the dashboard. Appendix D contains the central deployed rule logic and the owner email function.

Table 5.3 Script inventory

Location	Purpose	Deluge operations
Plans and person forms	Validate core values and duplicate names or emails	if, record criteria, count, alert
Memberships validation	Snapshot fee, derive expiry and total, reject overlap	fetch, addDay, sum, cancel submit
Memberships on load	Disable calculated fields and apply defaults	disable, hide, ifnull-style checks
Payments validation	Prevent invalid amount, status or date	sum excluding input.ID, conditions
Attendance and class rules	Validate timestamps, active eligibility and overlap	date criteria, count, fetch
Bookings validation	Prevent duplicate or ineligible reservations	count, capacity comparison
Send_Demo_Summary	Owner-only summary email	login guard, aggregate, sendmail, info

5.6 Workflows and Automation

Form workflows execute validation when records are created or edited. They reject invalid inputs before a record is saved. Membership defaults improve data entry but do not replace server-side validation. This version does not use approval workflows, schedules, Blueprints or batch workflows; its processes require immediate validation rather than staged approval.

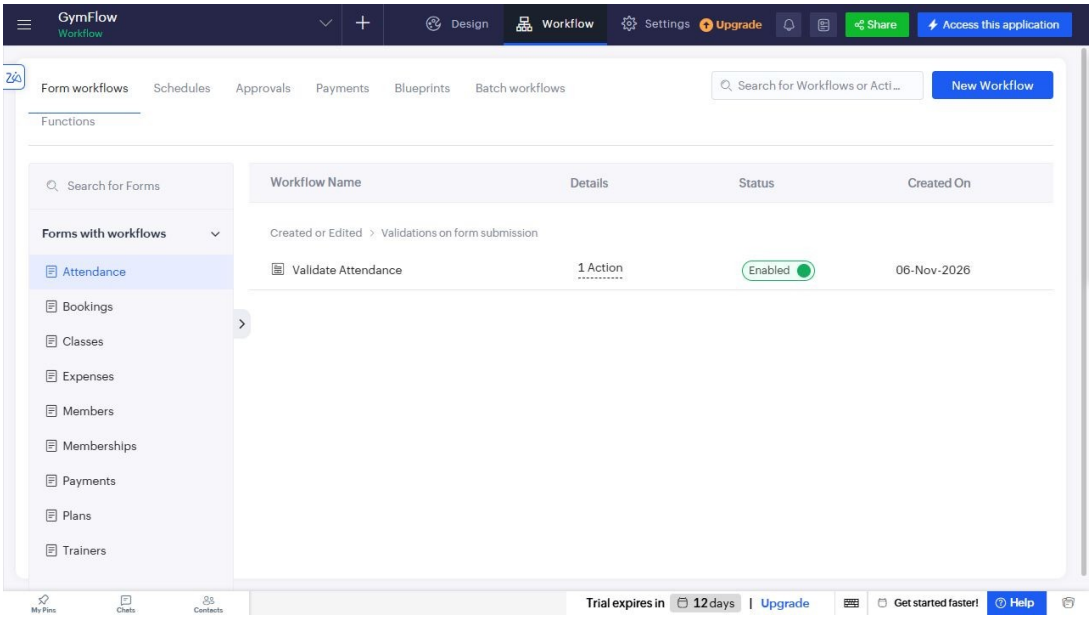


Figure 5.17 Enabled attendance validation workflow

5.7 Integrations

The integration demonstrated in this project is the native Creator sendmail task [4]. The Dashboard button calls Send_Demo_Summary, which permits execution only when the login email matches the administrator email. It calculates active memberships, month collections, expenses and net cash and sends a clearly labelled demonstration message to the owner. The subject is "GymFlow demo - live dashboard summary".

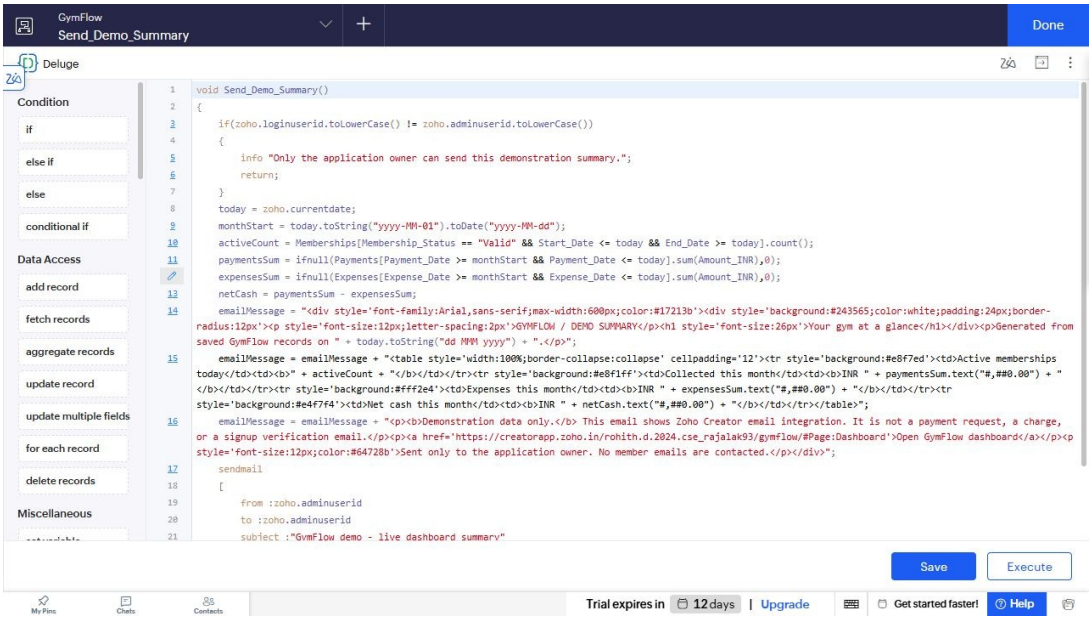


Figure 5.18 Reviewed owner email function in the Deluge editor

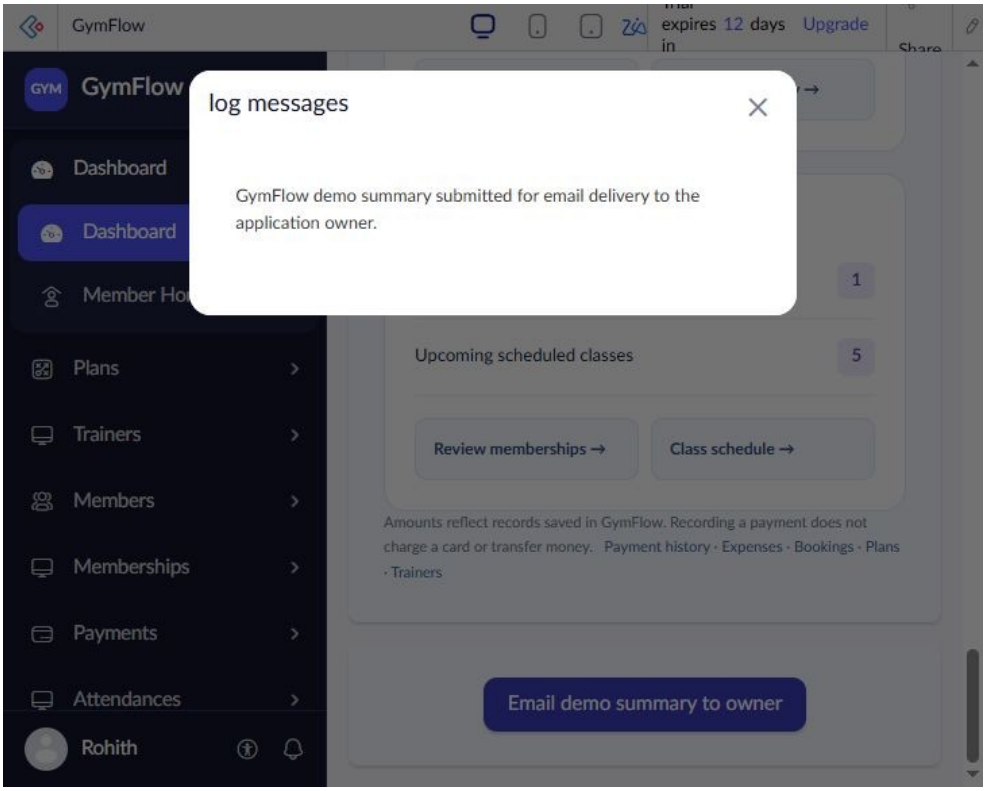


Figure 5.19 Successful dashboard email function submission

The owner confirmed receipt in the college inbox on 9 November 2026. This verifies notification delivery. It does not establish member signup verification, and it does not collect a payment. No external REST API, OAuth connection, payment gateway or Zoho CRM synchronization is configured.

5.8 User Management and Role Based Access

The owner account was used for configuration and staff-workflow demonstration. The intended portal profile should expose Member Home and only the signed-in member's records. However, attempts to save the narrowed portal permissions returned to the unchanged profile list. The report therefore records member access control as incomplete. Passwords are handled by Zoho and are not stored in GymFlow forms.

5.9 AI and Analytics Features

Zia Build Agent was used for a design review and generation of Send_Demo_Summary [5]. Its proposed chart modification could not retrieve the existing page structure, so the four charts were implemented directly. Review of the generated function identified a wrong status value and missing start-date condition. These were corrected, together with money formatting and the owner guard, before execution.

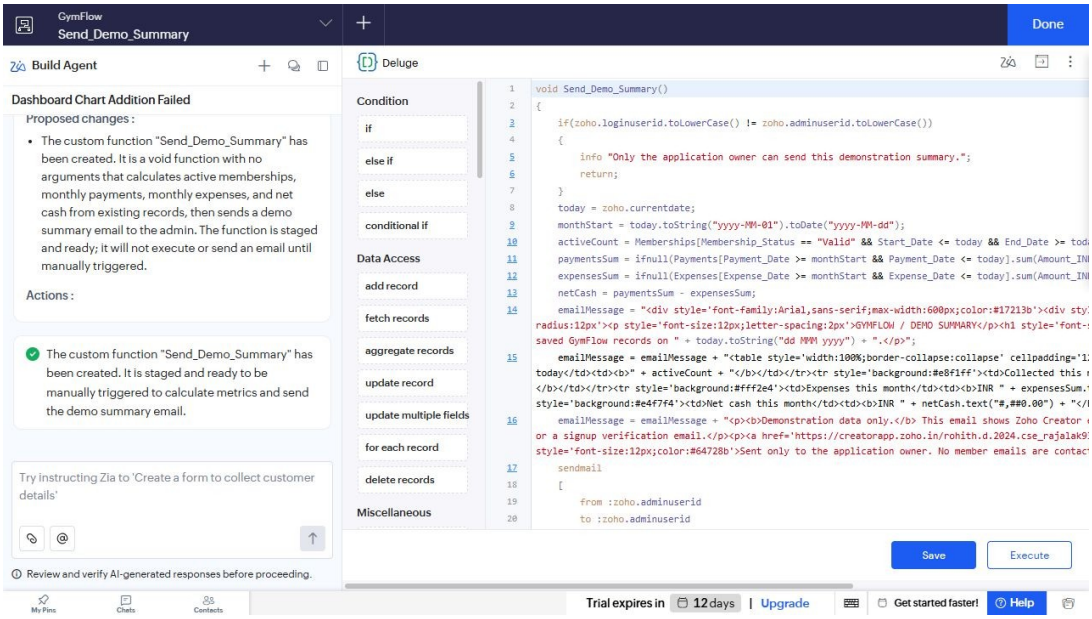


Figure 5.20 Zia function generation and the reviewed implementation

Analytics in this version summarize saved operational records. There is no trained prediction model, AI object-detection field or automated fitness recommendation. The AI contribution is development assistance, while the dashboard calculations are deterministic.

5.10 Pro Code Extension

The custom HTML/CSS dashboard and Deluge function extend the native visual components. No separate Catalyst service, Node.js backend, custom widget or Java function was deployed. This keeps the current application within the Creator environment.

5.11 Mobile and Portal Access

The live application provides web, phone and tablet preview controls. The dashboard layout responds to available width, but a separate native mobile device test is not claimed. A public portal signup page exists. Zoho does not permit external portal users to share the domain of the super administrator or internal users [6]; consequently, a separate external-domain inbox is needed to complete the member verification test. This test must follow correction of the member permission profile.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing used synthetic records, with five records in each of the nine forms. Positive tests checked saved records and calculated results; negative tests checked rejection of invalid submissions. Reports and dashboard totals were reconciled. Validation tests were performed during application construction, and dashboard, class dates, function execution and email receipt were checked again on 9 November 2026.

6.2 Test Cases

Table 6.1 Executed and pending tests

ID	Requirement and scenario	Observed result	Status
TC01	FR01 Invalid plan duration	Submission rejected	Pass
TC02	FR02 Create TEST membership	30-day expiry and INR 900 total calculated	Pass
TC03	FR03 Submit overpayment	Remaining-balance validation rejected input	Pass
TC04	FR03 Save partial payment	INR 300 receipt saved for TEST membership	Pass
TC05	FR04 Check in and check out	Visit and checkout saved	Pass
TC06	FR04 Duplicate open visit	Second open visit rejected	Pass
TC07	FR05 and FR06 Class and booking	Class and linked booking saved	Pass
TC08	FR06 Duplicate booking	Duplicate reservation rejected	Pass
TC09	FR02 Membership form behavior	Reference shown in payment selector; calculated fields disabled	Pass
TC10	FR07 Dashboard reconciliation	Collections 8600, expenses 6000, net cash 2600	Pass
TC11	FR08 Owner summary email	Function succeeded; user confirmed inbox receipt	Pass
TC12	FR09 Member-only access	Permission change not persisted	Pending
TC13	FR09 External signup and sign-in	External-inbox flow not completed	Pending
TC14	NFR Concurrent bookings and load	No multi-session load test performed	Not tested

6.3 Debugging Deluge Scripts

Validation messages identify the business rule that failed. The info statement reports whether the owner email request was submitted. The editor execution result and

dashboard button were both inspected. Function submission alone was not treated as delivery evidence; the inbox confirmation supplied that final check.

6.4 Defect Log

Table 6.2 Defects and resolutions

Issue	Correction or disposition	Status
Unclear payment lookup	Display Membership_Reference rather than only an internal identifier	Corrected
Calculated fields editable	Disable End_Date, Fee_INR and Total_INR; hide generated reference	Corrected
Zia used Active membership status	Use Valid and ensure the current date lies within the membership period	Corrected
Unformatted currency values	Apply grouped INR values with two decimal places	Corrected
Demo classes no longer upcoming	Move the five existing class dates to 11-12 November 2026	Corrected
Portal restriction save not persistent	Requires further configuration and a two-member access test	Open

6.5 User Acceptance Testing

The owner requested distinct demo data, a dashboard-first interface, colored indicators, charts and a college-email demonstration. These features were shown and retained. The owner explicitly confirmed receipt of the summary email. No independent staff acceptance survey or examiner approval is claimed. Member-facing acceptance remains pending until signup and access isolation are verified.

CHAPTER 7

DEPLOYMENT SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The demonstrated release runs in the existing GymFlow Creator application and is accessed through its live URL. Configuration and verification occurred in that application. A separate development-to-stage-to-production promotion was not performed. The account displayed a trial period, so licensing and continued access must be reviewed before ongoing operational use.

Table 7.1 Release readiness

Check	Current status
Application URL and owner access	Verified
Demonstration dataset	45 synthetic records retained deliberately
Dashboard and owner email	Verified
Member-only permissions	Pending persistent configuration
External member email signup	Pending separate inbox verification
Role and concurrency testing	Additional tests required
Independent production environment	Not established
Payment gateway or scheduled reminder	Not implemented

7.2 Performance Optimisation

Dashboard queries use criteria and aggregate functions for sums and counts. The chart period is limited to six months. The small dataset is suitable for a staff demonstration, but these choices are not evidence of large-scale performance. Before operational deployment, report load time, page-load query cost and simultaneous bookings should be measured with representative records.

7.3 Security Features

Creator sign-in controls access to the owner session. The email function checks the current login against the administrator before sending. Form validation rejects invalid business transactions. Sample member addresses use demonstration domains, and the summary is sent only to the owner. No passwords are recorded in the member data.

A member page filter must be supported by report and form permissions. Until the portal restrictions persist and direct-access tests succeed, the member portal is not

considered ready for real data. Tests must include one member attempting to access another member's records through both navigation and direct URLs.

7.4 Governance and Administration

Changes should be reviewed against the schema and tested before release. Keep a versioned copy of rules, page code and data definitions, and distinguish synthetic data from real records. Establish backup, record-retention and deletion procedures before daily use. Audit-trail operation, encryption configuration and restore recovery were not demonstrated and are not claimed as project results.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The verified dataset contains five plans, trainers, members, memberships, payments, attendance records, classes, bookings and expenses. All five memberships cover the verification date. Two attendance records remain open. Five scheduled classes are dated 11-12 November 2026. The following financial reconciliation uses the saved demonstration records.

Table 8.1 Verified financial reconciliation

Member	Membership total INR	Paid INR	Balance INR
TEST - GymFlow Member	900	300	600
Aarav Sharma	1500	1500	0
Diya Patel	1350	800	550
Vikram Singh	3500	3500	0
Nisha Reddy	5000	2500	2500
Total	12250	8600	3650

November collections of INR 8,600 minus expenses of INR 6,000 produce net cash of INR 2,600. This is a recorded cash-flow measure, not audited profit. Membership balances of INR 3,650 represent unpaid amounts; they are not automatically classified as overdue debt.

Table 8.2 Payment and expense composition

Payment method	Amount INR	Expense category	Amount INR
Cash	2800	Rent	2000
UPI	1500	Utilities	850
Card	800	Equipment	1200
Bank Transfer	3500	Maintenance	450
Total	8600	Salary	1500
		Expense total	6000

8.2 Achievement of Objectives

Table 8.3 Objectives and evidence

Objective	Result
Connected operational data	Nine forms, eight lookups and nine reports implemented.
Business validation	Membership, payment, attendance and booking checks demonstrated.
Management visibility	Six live indicators and four custom charts implemented.
Owner email integration	Dashboard function tested and college inbox receipt confirmed.
Member access isolation	Page filtering exists; portal permissions and external login remain incomplete.

8.3 Comparison with the Existing System

The proposed application links transactions to existing records and recalculates summaries from saved data, reducing the need for repeated manual matching. It provides immediate rejection of certain invalid entries. No measured reduction in processing time, staffing effort or cost is claimed because a controlled manual-versus-application study was not conducted.

8.4 Limitations of the Low Code Approach

The implementation depends on Creator availability, plan limits and its permission model. A generated AI proposal may contain incorrect schema assumptions. The dashboard must be refreshed after updates, and custom page charts do not provide every native analytical interaction. Concurrent capacity checks and deletion behavior need additional tests. These constraints define the next verification work rather than being hidden by the visual interface.

8.5 Reflection on the No Code Low Code and Pro Code Spectrum

Form layout, field configuration and list reports are mainly visual configuration. Deluge validation and owner notification are low-code logic. The custom HTML/CSS page requires direct coding within the platform. The project shows that these approaches can coexist, with code added where the business rules or presentation require it.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

GymFlow connects the main administrative activities of a gym in one Zoho Creator application. Its relationships avoid repeated member information, its workflows enforce operational rules, and its dashboard presents current totals from saved records. The owner email notification was executed and its receipt confirmed. The project is suitable for demonstrating the staff workflow with synthetic data, while member-portal security and external sign-in remain unfinished.

9.2 Future Enhancements

• Persist the member permission profile and verify isolated access using two real external test identities.

• Complete the signup, email-link, password-setup and sign-in sequence for an external member.

• Test concurrent bookings, report load and record deletion with a larger representative dataset.

• Introduce a separate development environment, backup checks and a controlled release process.

• Add expiry reminders, receipt templates or a payment gateway only after defining their operational requirements.

• Evaluate native analytical reports and mobile workflows where they improve daily staff use.

9.3 Learning Outcomes and Career Relevance

The project developed practical understanding of entity relationships, form design, validation events, date and currency calculations, report design and debugging. It also demonstrated the need to review AI-generated code and to distinguish page filtering from verified authorization. These skills apply to operational applications in other domains where transactions share common master data.

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[3] Zoho Corporation, "Gym fitness management software," Zoho Creator. Available: https://www.zoho.com/creator/apps/gym-fitness-management-software.html. Accessed: 9 November 2026.

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[6] Zoho Corporation, "Adding and managing portal users," Zoho Creator Help. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/portal/manage-and-configure-portal-users/articles/add-portal-users. Accessed: 9 November 2026.

APPENDIX A ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1 Implemented and unused platform features

Feature	Status	GymFlow evidence
Application and menu	Yes	Dashboard first; gym modules only
Eight or more field categories	Yes	Nine categories; Appendix C
Lookup relationships	Yes	Eight lookup fields
Subforms	No	Independent transaction forms used
Validation and field defaults	Yes	Form workflows and membership defaults
List reports	Yes	Nine native list reports
Kanban Calendar Timeline Map reports	No	Not required in the current demonstration
Native pivot or chart report	No	Four custom page charts used instead
Pages and dashboard	Yes	Dashboard and Member Home
Form-event Deluge	Yes	Rules across all nine forms
Functions fetch and aggregates	Yes	Send_Demo_Summary and page queries
Email notification	Yes	Owner delivery confirmed
Schedules approvals Blueprint batch	No	No such workflow configured
External REST API or webhook	No	No external API integration
User roles and portal restrictions	Partial	Owner session works; member boundary pending
AI assistance	Yes	Zia review and function draft
Analytics	Yes	Four deterministic custom charts
Measured performance benchmark	No	Only small demonstration dataset tested
Environment promotion	No	Single application demonstrated
Mobile device testing	No	Responsive layout only
Pro-code service or custom widget	No	HTML/CSS and Deluge inside Creator
Testing and debugging	Yes	Executed tests and open defects in Chapter 6

APPENDIX B COURSE OUTCOME MAPPING

Table B.1 Course outcome evidence

Outcome	Evidence in GymFlow	Extent
CO1 Explain low-code principles and architecture	Problem analysis, approach comparison and system architecture in Chapters 1-4	Demonstrated
CO2 Build forms reports workflows and Deluge logic	Nine forms and reports, validation rules and testing in Chapters 4-6	Demonstrated
CO3 Integrate services and apply role-based security	Owner email integration verified; portal security incomplete	Partial
CO4 Deploy secure and govern an application	Live single-app access and readiness review; staged deployment not demonstrated	Partial
CO5 Evaluate AI assistance and the development spectrum	Zia review, corrected code and custom dashboard discussion	Demonstrated

This mapping identifies project evidence and remaining gaps. It does not assign examiner marks or certify attainment levels.

APPENDIX C DATA DICTIONARY

The dictionary lists user-defined data fields. Creator system metadata such as record ID and added or modified timestamps is not included in the count of 48 fields. Lookup fields store record relationships even when the schema uses the picklist keyword. Phone values are Single Line fields; INR amounts are Decimal fields.

C.1 Plans

Table C.1 Plans field definitions

Link name	Type	Required	Meaning or constraint
Plan_Name	Single Line	Yes	Business rule described in Chapter 5
Duration_Days	Number	Yes	Business rule described in Chapter 5
Fee_INR	Decimal	Yes	Amount in INR
Plan_Status	Drop Down	Yes	Active, Inactive; Default Active
Benefits	Multi Line	No	Business rule described in Chapter 5

C.2 Trainers

Table C.2 Trainers field definitions

Link name	Type	Required	Meaning or constraint
Trainer_Name	Single Line	Yes	Business rule described in Chapter 5
Email	Email	Yes	Business rule described in Chapter 5
Phone	Single Line	No	Business rule described in Chapter 5
Specialization	Single Line	No	Business rule described in Chapter 5
Trainer_Status	Drop Down	Yes	Active, Inactive; Default Active

C.3 Members

Table C.3 Members field definitions

Link name	Type	Required	Meaning or constraint
Member_Name	Single Line	Yes	Business rule described in Chapter 5
Email	Email	Yes	Business rule described in Chapter 5
Phone	Single Line	No	Business rule described in Chapter 5
Joined_On	Date	Yes	Business rule described in Chapter 5
Assigned_Trainer	Lookup	No	Links Trainers
Member_Status	Drop Down	Yes	Active, Inactive; Default Active

C.4 Memberships

Table C.4 Memberships field definitions

Link name	Type	Required	Meaning or constraint
Member	Lookup	Yes	Links Members
Plan	Lookup	Yes	Links Plans
Start_Date	Date	Yes	Business rule described in Chapter 5
End_Date	Date	No	; calculated and disabled
Fee_INR	Decimal	No	Amount in INR; calculated and disabled
Discount_INR	Decimal	No	Amount in INR
Total_INR	Decimal	No	Amount in INR; calculated and disabled
Membership_Status	Drop Down	Yes	Valid, Cancelled; Default Valid
Membership_Reference	Single Line	No	Generated display reference; hidden in entry form

C.5 Payments

Table C.5 Payments field definitions

Link name	Type	Required	Meaning or constraint
Membership	Lookup	Yes	Links Memberships
Payment_Date	Date	Yes	Business rule described in Chapter 5
Amount_INR	Decimal	Yes	Amount in INR
Payment_Method	Drop Down	Yes	Cash, UPI, Card, Bank Transfer
Reference	Single Line	No	Business rule described in Chapter 5
Notes	Multi Line	No	Business rule described in Chapter 5

C.6 Attendance

Table C.6 Attendance field definitions

Link name	Type	Required	Meaning or constraint
Member	Lookup	Yes	Links Members
Check_In	Date-Time	Yes	Business rule described in Chapter 5
Check_Out	Date-Time	No	Blank while the visit is open

C.7 Classes

Table C.7 Classes field definitions

Link name	Type	Required	Meaning or constraint
Class_Name	Single Line	Yes	Business rule described in Chapter 5
Trainer	Lookup	Yes	Links Trainers
Starts_At	Date-Time	Yes	Business rule described in Chapter 5
Ends_At	Date-Time	Yes	Business rule described in Chapter 5
Capacity	Number	Yes	Business rule described in Chapter 5
Class_Status	Drop Down	Yes	Scheduled, Completed, Cancelled; Default Scheduled

C.8 Bookings

Table C.8 Bookings field definitions

Link name	Type	Required	Meaning or constraint
Member	Lookup	Yes	Links Members
Gym_Class	Lookup	Yes	Links Classes
Booking_Status	Drop Down	Yes	Confirmed, Cancelled, Attended, No Show; Default Confirmed

C.9 Expenses

Table C.9 Expenses field definitions

Link name	Type	Required	Meaning or constraint
Expense_Date	Date	Yes	Business rule described in Chapter 5
Category	Drop Down	Yes	Rent, Utilities, Equipment, Maintenance, Salary, Other
Amount_INR	Decimal	Yes	Amount in INR
Description	Multi Line	Yes	Business rule described in Chapter 5
Reference	Single Line	No	Business rule described in Chapter 5

APPENDIX D DELUGE SOURCE CODE

The following source records the central membership, payment and booking rules, membership field defaults and the owner summary function. Additional validation rules for plans, people, attendance, classes and expenses are described in Chapters 4-6. Display line wraps do not change the underlying logic.

D.1 Membership validation

```
planRecord = Plans[ID == input.Plan];
memberRecord = Members[ID == input.Member];
input.Membership_Reference = memberRecord.Member_Name + " / " +
planRecord.Plan_Name + " / " + input.Start_Date.toString("dd-MMM-
yyyy");
if(memberRecord.Member_Status != "Active")
{
 alert "Activate the member before creating a membership.";
 cancel submit;
}
oldRecord = Memberships[ID == input.ID];
if(oldRecord.count() == 0)
{
 if(planRecord.Plan_Status != "Active")
 {
  alert "Select an active membership plan.";
  cancel submit;
 }
 input.Fee_INR = planRecord.Fee_INR;
 input.End_Date = input.Start_Date.addDay(planRecord.Duration_Days -
1);
}
else
{
 if(oldRecord.Member != input.Member || oldRecord.Plan != input.Plan
|| oldRecord.Start_Date != input.Start_Date)
 {
  alert "Create a new membership for a renewal or plan change.
Existing member, plan and start date cannot be changed.";
  cancel submit;
 }
 input.Fee_INR = oldRecord.Fee_INR;
 input.End_Date = oldRecord.End_Date;
}
input.Discount_INR = ifnull(input.Discount_INR,0);
if(input.Discount_INR < 0 || input.Discount_INR > input.Fee_INR)
{
 alert "Discount must be between zero and the plan fee.";
 cancel submit;
}
input.Total_INR = input.Fee_INR - input.Discount_INR;
if(ifnull(Payments[Membership == input.ID].sum(Amount_INR),0) >
input.Total_INR)
{
 alert "The membership total cannot be less than payments already
received.";
 cancel submit;
}
if(input.Membership_Status == "Valid" && Memberships[Member ==
input.Member && ID != input.ID && Membership_Status == "Valid" &&
```

```
Start_Date <= input.End_Date && End_Date >= input.Start_Date].count()
> 0)
{
 alert "This member already has a membership covering these dates.";
 cancel submit;
}
```

D.2 Payment validation

```
membershipRecord = Memberships[ID == input.Membership];
if(membershipRecord.Membership_Status != "Valid")
{
 alert "Payments cannot be recorded for a cancelled membership.";
 cancel submit;
}
paid = ifnull(Payments[Membership == input.Membership && ID !=
input.ID].sum(Amount_INR),0);
if(input.Amount_INR <= 0 || paid + input.Amount_INR >
membershipRecord.Total_INR)
{
 alert "Enter a positive payment no greater than the remaining
balance: INR " + (membershipRecord.Total_INR - paid);
 cancel submit;
}
if(input.Payment_Date > zoho.currentdate)
{
 alert "Payment date cannot be in the future.";
 cancel submit;
}
```

D.3 Booking validation

```
if(input.Booking_Status != "Cancelled")
{
 classRecord = Classes[ID == input.Gym_Class];
 classDate = classRecord.Starts_At.toDate();
 memberRecord = Members[ID == input.Member];
 if(classRecord.Class_Status == "Cancelled")
 {
  alert "This class has been cancelled.";
  cancel submit;
 }
 if(memberRecord.Member_Status != "Active" || Memberships[Member ==
input.Member && Membership_Status == "Valid" && Start_Date <=
classDate && End_Date >= classDate].count() == 0)
 {
  alert "An active membership covering the class date is required.";
  cancel submit;
 }
 if(Bookings[Member == input.Member && Gym_Class == input.Gym_Class
&& ID != input.ID && Booking_Status != "Cancelled"].count() > 0)
 {
  alert "This member is already booked into this class.";
  cancel submit;
 }
 if(Bookings[Gym_Class == input.Gym_Class && ID != input.ID &&
Booking_Status != "Cancelled"].count() >= classRecord.Capacity)
 {
  alert "This class is full.";
  cancel submit;
 }
 if(input.Booking_Status == "Confirmed" &&
```

```
(classRecord.Class_Status != "Scheduled" || classRecord.Starts_At <=
zoho.currenttime))
 {
  alert "Bookings are available only for upcoming scheduled
classes.";
  cancel submit;
 }
}
```

D.4 Membership form defaults

```
disable End_Date;
disable Fee_INR;
disable Total_INR;
hide Membership_Reference;
if(input.Start_Date == null)
{
 input.Start_Date = zoho.currentdate;
}
if(input.Discount_INR == null)
{
 input.Discount_INR = 0;
}
```

D.5 Owner summary email

```
void Send_Demo_Summary()
{
    if(zoho.loginuserid.toLowerCase() !=
zoho.adminuserid.toLowerCase())
    {
        info "Only the application owner can send this demonstration
summary.";
        return;
    }
    today = zoho.currentdate;
    monthStart = today.toString("yyyy-MM-01").toDate("yyyy-MM-dd");
    activeCount = Memberships[Membership_Status == "Valid" &&
Start_Date <= today && End_Date >= today].count();
    paymentsSum = ifnull(Payments[Payment_Date >= monthStart &&
Payment_Date <= today].sum(Amount_INR),0);
    expensesSum = ifnull(Expenses[Expense_Date >= monthStart &&
Expense_Date <= today].sum(Amount_INR),0);
    netCash = paymentsSum - expensesSum;
    emailMessage = "<div style='font-family:Arial,sans-serif;max-
width:600px;color:#17213b'><div
style='background:#243565;color:white;padding:24px;border-
radius:12px'><p style='font-size:12px;letter-spacing:2px'>GYMFLOW /
DEMO SUMMARY</p><h1 style='font-size:26px'>Your gym at a
glance</h1></div><p>Generated from saved GymFlow records on " +
today.toString("dd MMM yyyy") + ".</p>";
    emailMessage = emailMessage + "<table style='width:100%;border-
collapse:collapse' cellpadding='12'><tr
style='background:#e8f7ed'><td>Active memberships today</td><td><b>"
+ activeCount + "</b></td></tr><tr
style='background:#e8f1ff'><td>Collected this month</td><td><b>INR "
+ paymentsSum.text("#,##0.00") + "</b></td></tr><tr
style='background:#fff2e4'><td>Expenses this month</td><td><b>INR " +
expensesSum.text("#,##0.00") + "</b></td></tr><tr
style='background:#e4f7f4'><td>Net cash this month</td><td><b>INR " +
netCash.text("#,##0.00") + "</b></td></tr></table>";
    emailMessage = emailMessage + "<p><b>Demonstration data only.</b>
```

```
This email shows Zoho Creator email integration. It is not a payment
request, a charge, or a signup verification email.</p><p><a
href='https://creatorapp.zoho.in/rohith.d.2024.cse_rajalak93/gymflow/
#Page:Dashboard'>Open GymFlow dashboard</a></p><p style='font-
size:12px;color:#64728b'>Sent only to the application owner. No
member emails are contacted.</p></div>";
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"GymFlow demo - live dashboard summary"
        message :emailMessage
        content type :HTML
    ]
    info "GymFlow demo summary submitted for email delivery to the
application owner.";
}
```

APPENDIX E USER MANUAL

E.1 Accessing the Application

Sign in to the owner Zoho account and open the GymFlow live URL from the project profile. Dashboard is the landing page. The figures are calculated when the page loads; refresh after editing records.

E.2 Registering Members and Memberships

Create an active plan and trainer if required. Add a member with a unique email, joining date and optional trainer. Open Memberships, select the member and plan, enter the start date and discount, and save. The application generates the expiry, fee, total and readable reference. For renewals, create a new membership period.

E.3 Recording Payments and Attendance

Open Payments, choose the membership reference, and enter date, positive amount and method. The amount must not exceed the outstanding balance. To record a visit, open Attendance, select the member and enter check-in. Update the same visit with check-out when it ends; do not create another open visit for the same member.

E.4 Classes Bookings and Expenses

Create a class with an active trainer, future time interval, positive capacity and Scheduled status. Create a booking by selecting the member and class. The member must have a valid membership on the class date. Record operating costs in Expenses with the appropriate category and a positive amount.

E.5 Demonstrating the Dashboard and Email

Show the six dashboard indicators, then the four charts. Scroll to the email button and click it once while signed in as the owner. Close the submission log and check the college inbox for "GymFlow demo - live dashboard summary". The received message demonstrates notification integration; it is not a signup verification email.

E.6 Portal Verification Before Member Use

First complete the member-only permission configuration. Then use an external-domain inbox owned by the tester, complete signup and CAPTCHA, open Zoho's verification/password-setup link, set a password and sign in. Match the verified email to the corresponding member record. Test two identities to confirm that each can see only its own data. This procedure remains future verification work.

APPENDIX F EVALUATION RUBRIC

Student: Rohith D Register number: 2116240701435

Project: GymFlow Gym Management System

Table F.1 Individual project evaluation

No	Criterion	Maximum	Awarded
1	Problem definition and requirement analysis	10	
2	System design	10	
3	Forms reports and pages	15	
4	Deluge scripting and workflows	15	
5	Integration RBAC and AI	10	
6	Testing deployment security and governance	10	
7	Live demonstration	15	
8	Project report	5	
9	Viva voce	10	
	TOTAL	100	

Date: _____

Internal examiner signature: _____

External examiner signature: _____

APPENDIX G VIVA VOCE RECORD SHEET

Student: Rohith D Register number: 2116240701435
Department: Computer Science and Engineering
Project: GymFlow Gym Management System

Date of viva: _____ Demo duration: _____ minutes
Application URL verified: Yes / No

Table G.1 Questions and responses

No	Question or topic	Unit	Response	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

Response key: E - Excellent; G - Good; S - Satisfactory; NI - Needs improvement.

Table G.2 Marks summary

Component	Maximum	Awarded
Criteria 1 to 8 in Appendix F	90	
Viva voce	10	
Total	100	

Examiner remarks: _____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Signature: _____ Signature: _____